

# **A Video-Based Object Tracking Framework using ViBe and Kalman Filter**

## **Project Report**

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### **1. Introduction**

This project is based on a master's thesis completed in January 2019. The code and documentation were reorganized and prepared as a GitHub portfolio project in 2026.

Information obtained from images and videos plays a vital role in the detection and analysis of moving objects. Computer vision enables computers to obtain visual information, process it, make decisions, and generate the appropriate commands for control systems. Object detection and tracking are among the most important topics in computer vision and have been widely applied in various fields, including industry, medicine, military systems, security, and video surveillance. These technologies are also used in advanced driver assistance systems (ADAS) for obstacle detection and collision avoidance, surrounding object detection to assist visually impaired individuals, person identification, and many other real-world applications.

Despite significant progress in this field, object detection and tracking still face several challenges. A large number of objects, complex backgrounds, camera viewpoint changes, varying illumination conditions, object pose variations, and camera characteristics can all reduce tracking accuracy.

A typical computer vision system begins by capturing images through a camera and then extracting the required information using digital image processing techniques. Common processing operations include image enhancement, image restoration and fusion, filtering, thresholding, edge detection, pattern recognition, color analysis, and feature extraction. The extracted features should not only provide effective object representation but also be robust against scale and rotation changes, thereby improving classification accuracy.

The performance of an object detection system largely depends on selecting appropriate feature extraction and machine learning methods according to the target application. For example, in video surveillance systems, computer vision is employed to recognize activities and generate alarms when necessary. In intelligent transportation systems, the primary focus is on vehicle tracking and accident detection. Therefore, correctly defining the problem and selecting suitable methods play a crucial role in designing an efficient computer vision system.

Considering these challenges, improving the robustness of object tracking algorithms under difficult conditions, particularly during partial or complete occlusions, remains an important research topic in computer vision. Furthermore, developing methods that improve accuracy while maintaining lower computational complexity is essential for real-time applications.

## **2. Research Objective**

This study aims to optimize the classical ViBe algorithm by reducing processing time and improving its performance under challenging conditions. In addition, the proposed algorithm was implemented and experimentally evaluated. To evaluate the proposed algorithm, the Euclidean distance between the estimated object center and the Ground Truth coordinates was compared with the Euclidean distance obtained using the Mean Shift algorithm. Furthermore, the estimated x- and y-coordinates were compared with the Ground Truth coordinates to analyze localization accuracy. The processing speed of the proposed algorithm was also compared with both the classical ViBe algorithm and the Mean Shift algorithm.

Since the classical ViBe algorithm performs poorly under partial and complete occlusions, a Kalman filter was integrated with the original algorithm to overcome this limitation. Experimental results demonstrate that the proposed algorithm is more robust under challenging conditions, especially during occlusions, while also processing faster than the classical ViBe algorithm.

## **3. Proposed Method**

In this study, the Visual Background Extractor (ViBe) algorithm was adopted as the baseline method. Since the ViBe background subtraction algorithm has limited robustness against noise, a morphological filter was applied to remove noise generated by the foreground extraction process. Subsequently, a Kalman filter was used to enhance the tracking performance.

The Kalman filter predicts the future trajectory of the target and estimates its approximate position during partial or complete occlusions, preventing the system from losing the object when it temporarily disappears from the scene. As a recursive estimation technique, the Kalman filter efficiently combines information in dynamic and uncertain environments without requiring storage of all previous observations, making it particularly suitable for real-time systems.

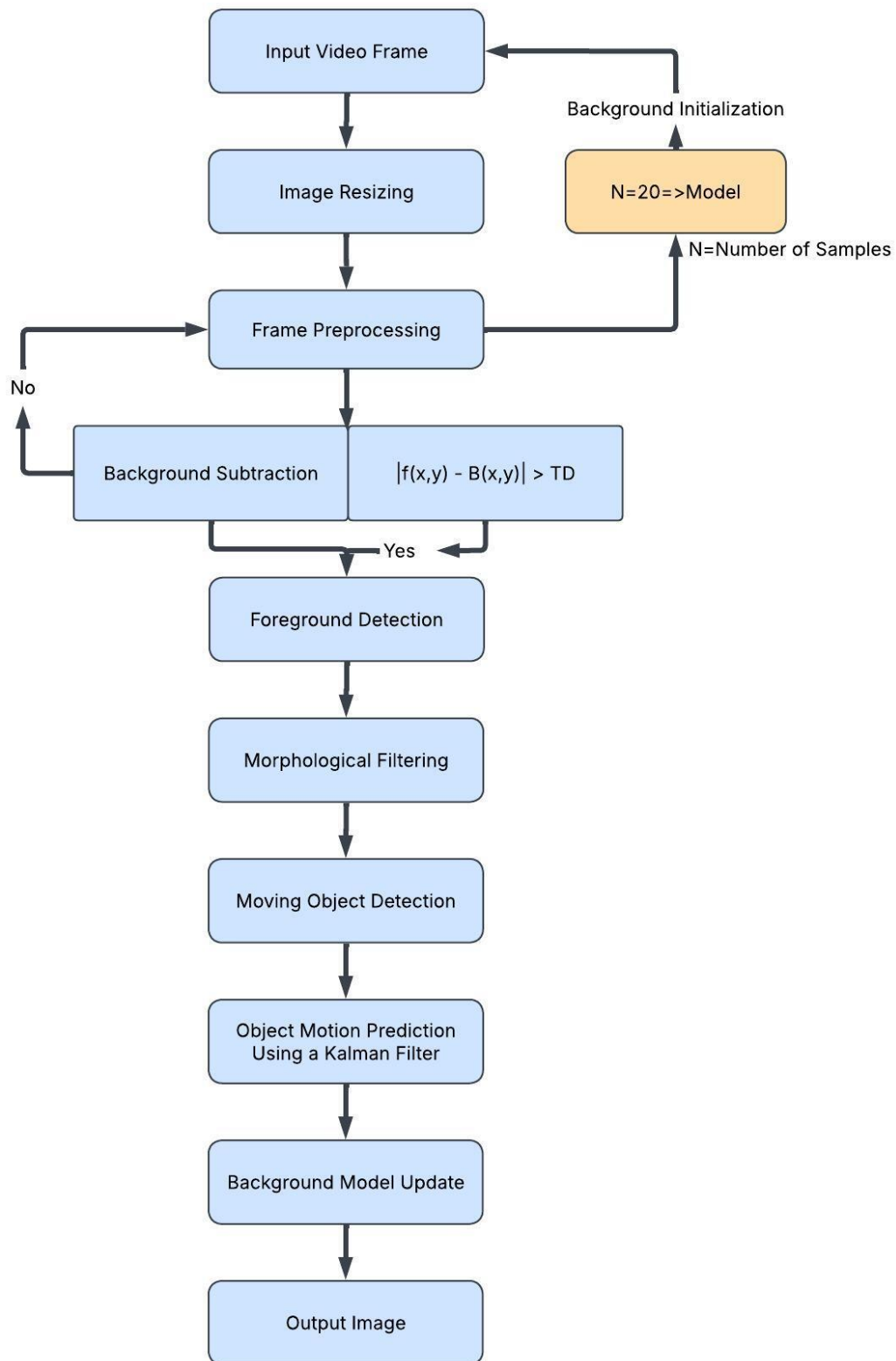


Figure 3.1. Flowchart of the proposed algorithm.

Figure 3.1 provides an overview of the proposed algorithm. A detailed explanation of each processing stage is provided below.

The proposed algorithm first takes a video as input, decomposes it into individual frames, and constructs a background model. By comparing the background model with each incoming frame using a predefined threshold, moving objects are detected. Morphological filtering is then employed to remove noise and shadows caused by false detections. Next, the Kalman filter is applied to overcome the limitations of the classical algorithm, particularly during occlusions, and to maintain stable object tracking. Finally, the background model is updated, and the processed output frame is generated.

**Algorithm 1: ViBe-Based  
Background Subtraction with  
Kalman Filter Tracking**

Input: Video sequence

Output: Tracked object position

1. Initialize background model
2. For each frame:
  - a. Convert frame to grayscale
  - b. Apply ViBe segmentation
  - c. Apply morphological filtering
  - d. Extract object centroid
  - e. Kalman filter prediction and update
3. Return tracking results

Algorithm 1 shows the proposed tracking pipeline based on ViBe and Kalman filter.

By combining the classical ViBe algorithm with a Kalman filter, the proposed method enhances both foreground detection and object tracking performance. The ViBe algorithm accurately models the background and detects moving objects, while the

Kalman filter predicts the object's future position, improving tracking stability. As a result, the proposed algorithm is capable of maintaining accurate tracking even under challenging conditions where conventional methods may lose the target.

To evaluate the proposed algorithm, the Mean Shift tracking algorithm was selected as the baseline comparison method.

The Mean Shift tracker relies on color histogram information and uses three color channels for tracking. Therefore, at least 50% of the target must remain visible in each frame for successful tracking. Consequently, it performs poorly under complete occlusion and loses the target, although it generally performs well under partial occlusion. Furthermore, the object must already be present in the scene during initialization, as Mean Shift cannot automatically detect objects entering from outside the observed scene.

The proposed algorithm is computationally more efficient because it operates effectively on grayscale images, making it more suitable for real-time applications. Moreover, it can estimate the target location under both partial and complete occlusions and is capable of automatically detecting and tracking moving objects after learning the background from the initial frames.

#### 4. Experimental Setup

The proposed algorithm was implemented in MATLAB and tested using three publicly available video sequences: **Box**, **car-overhead-1**, and **car-overhead-2-hires**. To maintain consistency throughout this report, these datasets are referred to as **Box**, **Car1**, and **Car2**, respectively.

The Ground Truth annotations provided with each dataset were used as the reference for evaluation. These annotations consist of the bounding box coordinates of the target object in each frame.

The algorithm's performance was evaluated by comparing the estimated object center with the corresponding Ground Truth coordinates. The Euclidean distance between these two positions was calculated for every frame to measure localization accuracy.

In addition, the estimated x- and y-coordinates were compared with the corresponding Ground Truth coordinates to analyze the tracking trajectory and localization error.

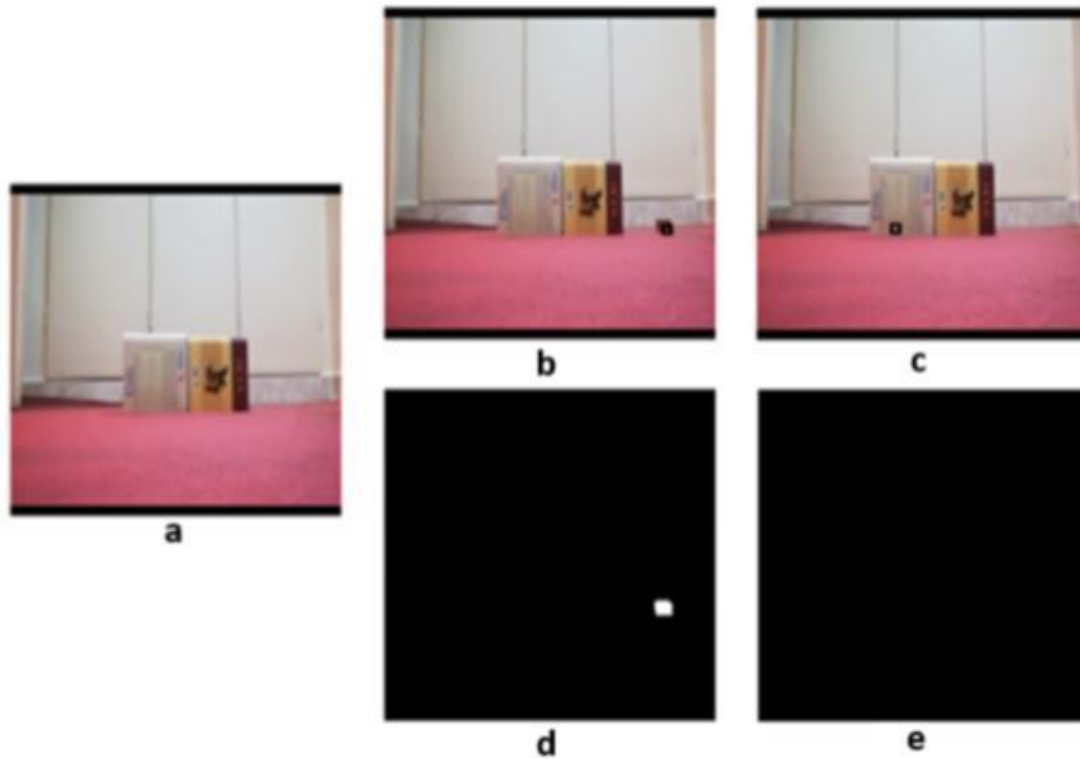


Figure 4-1. Results of the proposed algorithm on the Box video. (a) Background image without the moving object, (b) moving foreground object, (c) occlusion scenario with the object position estimated by the Kalman filter, (d) ground-truth tracking annotation corresponding to (b), and (e) ground-truth tracking annotation corresponding to the occlusion scenario in (c).

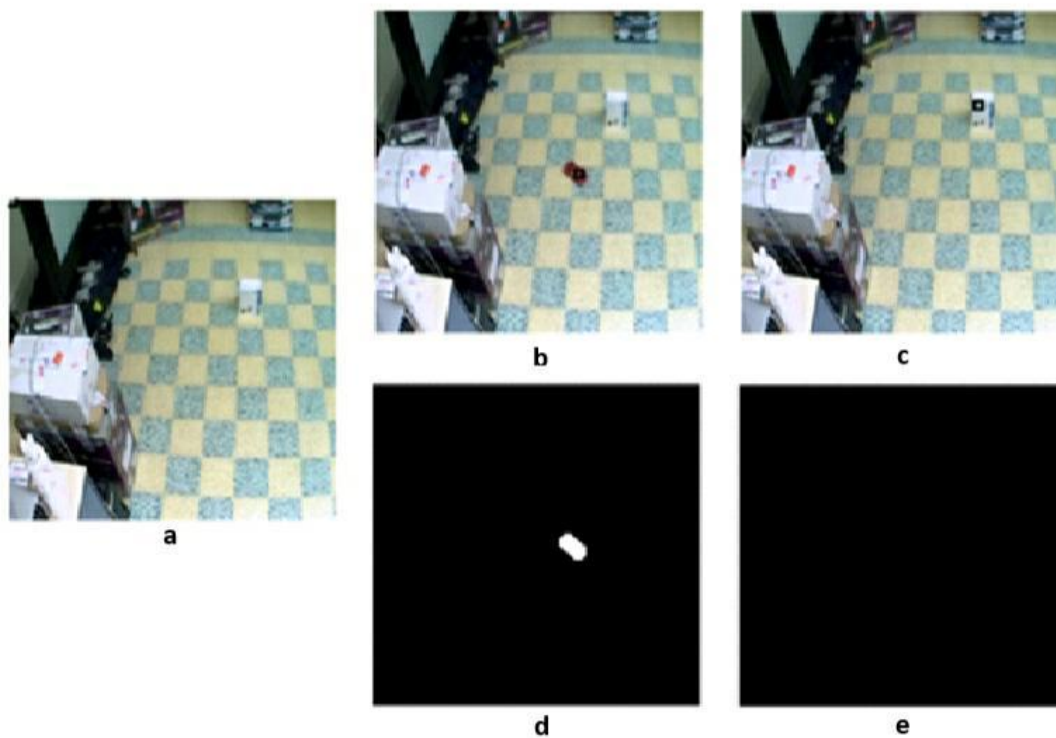


Figure 4-2. Results of the proposed algorithm on the Car1 video: (a) background image, (b) moving foreground object, (c) occlusion with Kalman filter prediction, (d) ground-truth tracking annotation corresponding to (b), and (e) ground-truth tracking annotation corresponding to the occlusion scenario in (c).

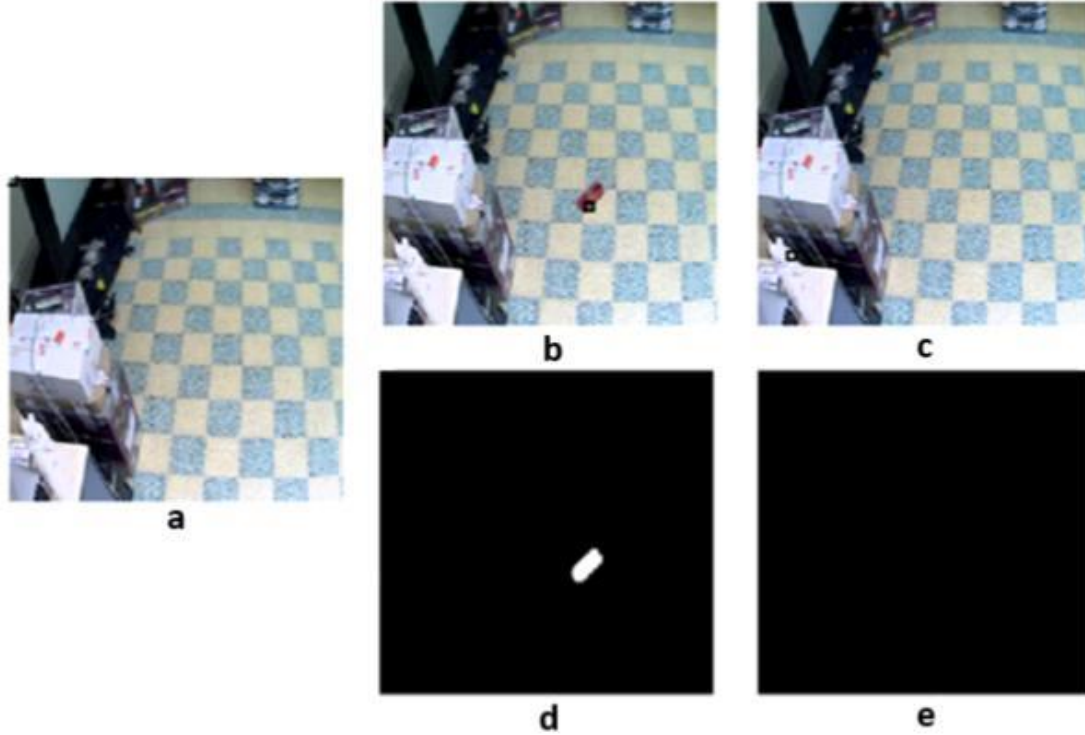


Figure 4-3. Results of the proposed algorithm on the Car2 video: (a) background image, (b) moving foreground object, (c) occlusion with Kalman filter prediction, (d) ground-truth tracking annotation corresponding to (b), and (e) ground-truth tracking annotation corresponding to the occlusion scenario in (c).

## 5. Results and Discussion

The proposed algorithm was evaluated against the Mean Shift tracking algorithm using the Euclidean distance metric. Experimental results indicate that the proposed algorithm performs better than Mean Shift in challenging scenarios, particularly in the Box sequence, while achieving comparable performance in the remaining datasets. Furthermore, the x- and y-coordinate error plots demonstrate that the proposed algorithm accurately estimates the object position.

The use of the Kalman filter significantly improves tracking stability. In addition, combining Kalman filtering with background subtraction provides an effective balance between tracking accuracy and computational efficiency, enabling more robust performance under challenging conditions.

According to the experimental results, the proposed algorithm converges to the moving target within an average of 26 frames (approximately 0.86 seconds) after the object enters the scene. In the Box sequence, the initial localization error decreases from 218 to 134 before converging to the target. In Car1, the error decreases from 300 to 112, while in Car2, it decreases from 637 to 234 before convergence.



In contrast, the Mean Shift algorithm requires the target location to be manually initialized in the first frame and depends on the target being initially visible in the scene. Therefore, it cannot automatically detect newly appearing objects.

In all plots, the green curves represent the proposed algorithm, the blue curves correspond to the Mean Shift algorithm, and the red curves indicate the Ground Truth. The first plots illustrate the x-coordinate error, the second plots show the y-coordinate error, and the final plots present the Euclidean distance error.

Figure 5.1 shows the results for the Box sequence. The Box sequence contains a long complete occlusion. The proposed algorithm successfully estimates the object position during the occlusion period, whereas the Mean Shift algorithm loses the target during complete occlusion and fails to recover it after the object reappears.

Figure 5.2 presents the results for the Car1 sequence. The Car1 sequence contains several partial and non-consecutive occlusions. Both algorithms successfully track the target under these conditions. Since Mean Shift is based on color histograms and requires a portion of the object to remain visible, it performs well during partial occlusions.

Figure 5.3 illustrates the results for the Car2 sequence. The Car2 sequence presents a simple scenario in which the vehicle enters from the top of the frame and exits from one corner. According to the results, both algorithms successfully track the target

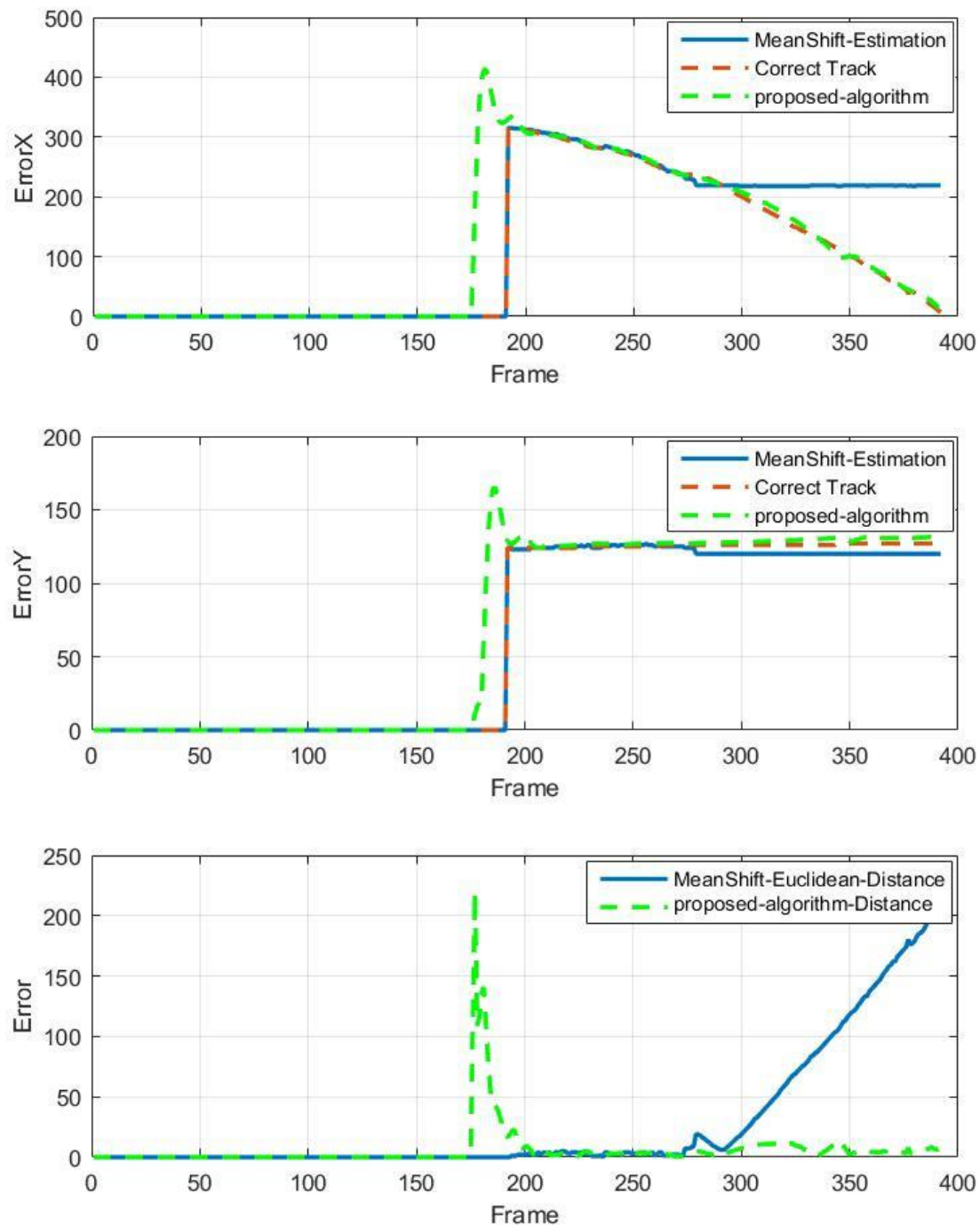


Figure 5-1. Performance evaluation of the proposed tracking algorithm on the Box video. (a) Comparison of the estimated X-coordinate with the ground-truth trajectory and Mean Shift tracker, (b) comparison of the estimated Y-coordinate with the ground-truth trajectory and Mean Shift tracker, and (c) Euclidean distance between the estimated target position and the corresponding ground truth for the proposed algorithm and the conventional Mean Shift tracker.

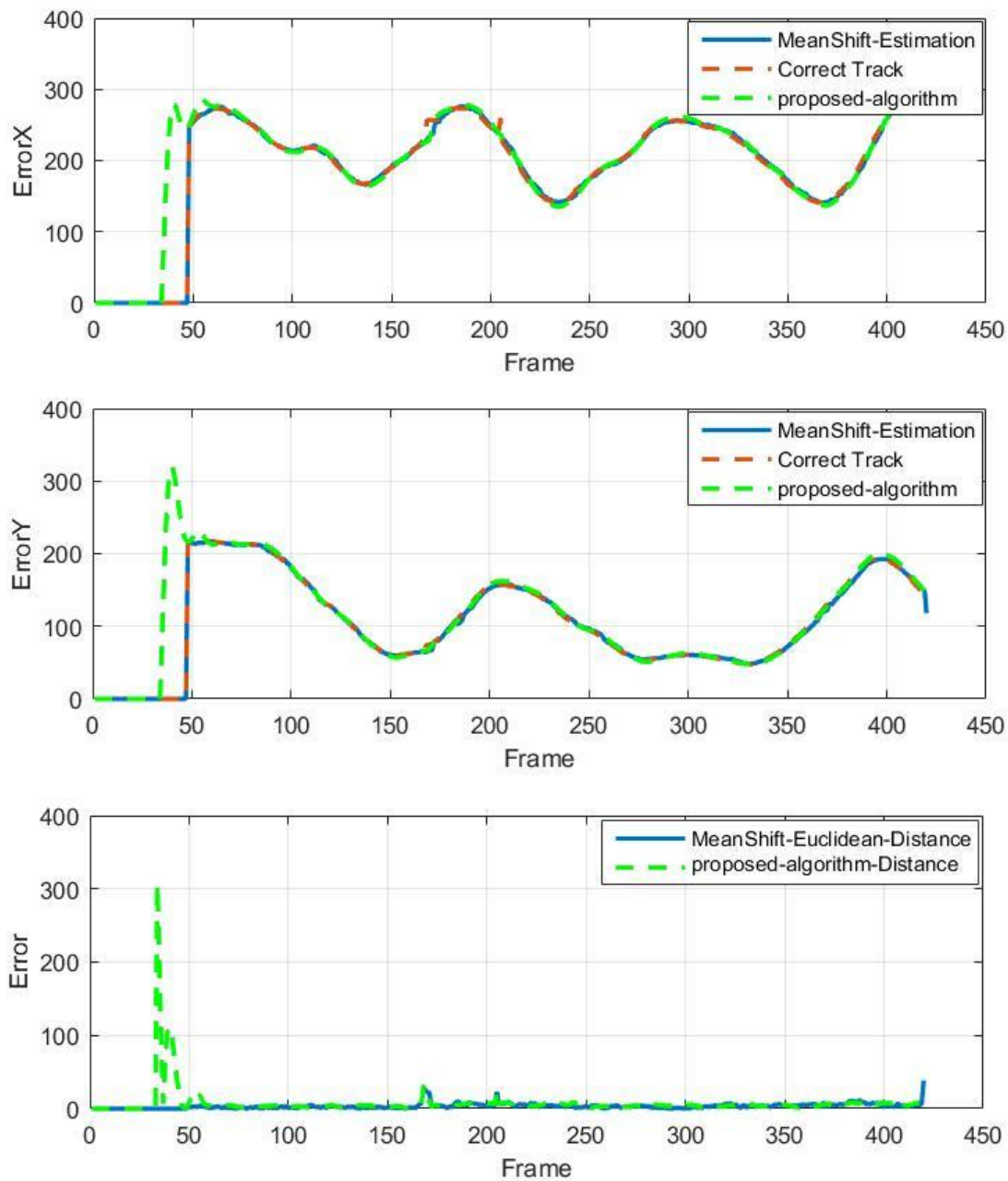


Figure 5-2. Performance evaluation of the proposed tracking algorithm on the Car1 video. (a) Comparison of the estimated X-coordinate with the ground-truth trajectory and Mean Shift tracker, (b) comparison of the estimated Y-coordinate with the ground-truth trajectory and Mean Shift tracker, and (c) Euclidean distance between the estimated target position and the corresponding ground truth for the proposed algorithm and the conventional Mean Shift tracker.

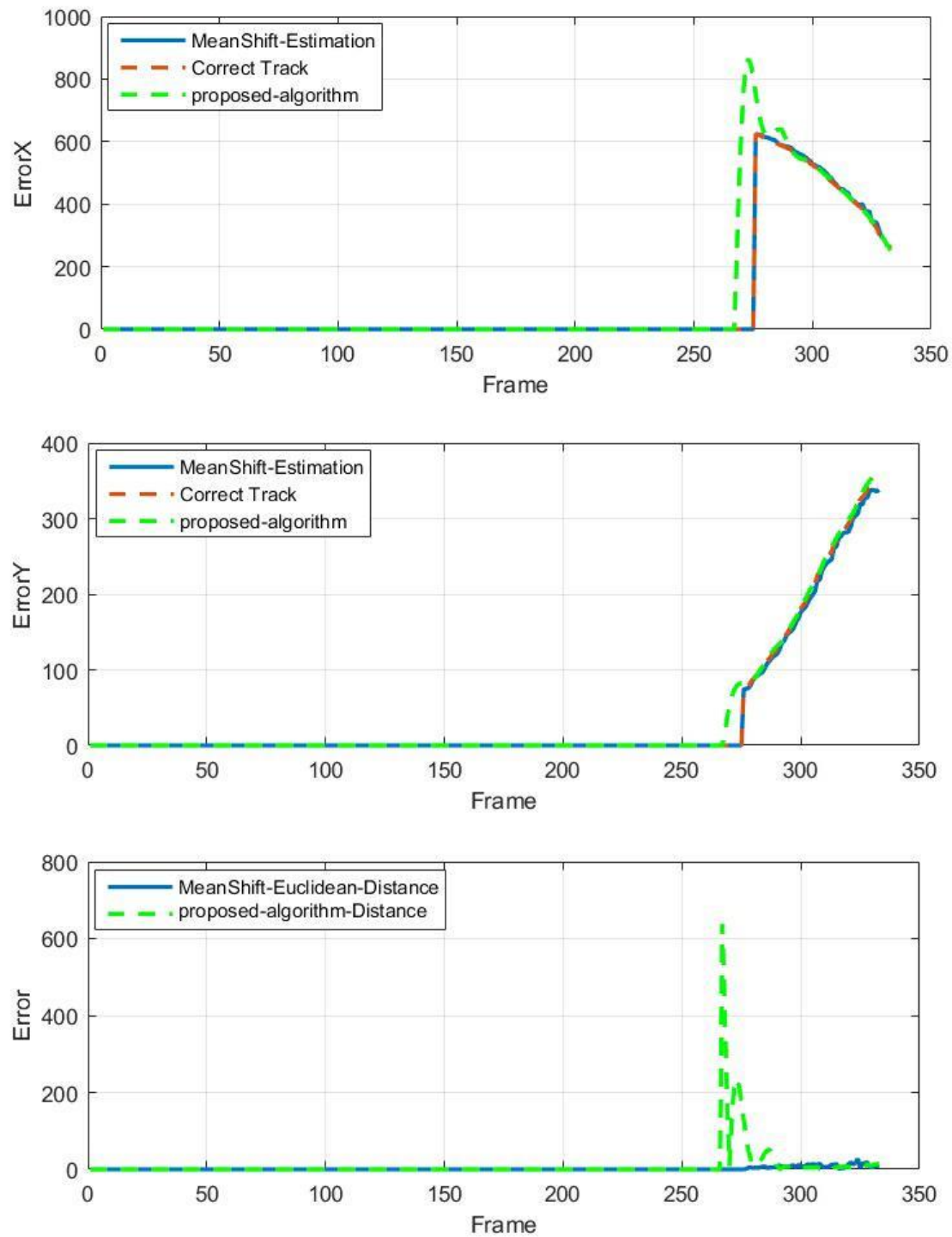


Figure 5-3. Performance evaluation of the proposed tracking algorithm on the Car2 video. (a) Comparison of the estimated X-coordinate with the ground-truth trajectory and Mean Shift tracker, (b) comparison of the estimated Y-coordinate with the ground-truth trajectory and Mean Shift tracker, and (c) Euclidean distance between the

estimated target position and the corresponding ground truth for the proposed algorithm and the conventional Mean Shift tracker.

| Video's name | Processing time of the ViBe algorithm | Processing time of the proposed algorithm | Processing time of the Mean Shift algorithm |
|--------------|---------------------------------------|-------------------------------------------|---------------------------------------------|
| Box          | ~0.12 Sec                             | ~0.10 Sec                                 | ~0.15 Sec                                   |
| Car1         | ~0.12 Sec                             | ~0.009 Sec                                | ~0.10 Sec                                   |
| Car2         | ~0.13 Sec                             | ~0.14 Sec                                 | ~0.15 Sec                                   |

Table 5-1. Processing time comparison among the classical ViBe algorithm, the proposed algorithm, and the Mean Shift tracker.

The processing times of the classical ViBe algorithm, the proposed algorithm, and the Mean Shift algorithm were compared. The experimental results demonstrate that the proposed algorithm achieves the highest processing speed among the three algorithms.

## 6. Contributions

This research proposes an improved object tracking framework by integrating the classical ViBe algorithm with a Kalman filter to enhance localization and tracking performance. The proposed algorithm was quantitatively evaluated using Ground Truth annotations and the Euclidean distance metric. In addition, its computational efficiency was compared with both the classical ViBe algorithm and the Mean Shift tracker, demonstrating faster processing while maintaining robust tracking performance.

## 7. Future Work

Future research may focus on improving the algorithm's robustness against image noise and extending its applicability to videos captured by moving cameras. Adaptive parameter and threshold selection techniques could further improve tracking performance. Another possible direction is to extend the framework to detect and track multiple moving objects in crowded scenes. In addition, investigating methods to improve robustness against scale variations represents an important area for future research.